

Coordinating Tasks in M-Form and U-Form Organisa

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Introduction

Introduction

- Coordination becomes crucial whenever there is specialization.
- Example: Harvesting and transport/storage.
- It is like assembling complementary parts. Each complementary part is characterized by its attributes in dimensions.
- In reforms or innovations, it is not sufficient to match all attributes in blueprints.
- Blueprints of reform are decided by the top manager.
- We want to use this model to examine how alternative organizational forms affect coordination.

Preparations For Modelling

Preparations For Modelling

- We first consider an organization with two regions “A” and “B” and two functions “1” and “2”. Assume an infinite time horizon.
- Prior to any reform, the existing technology generates payoffs of $\frac{R}{2}$ in every period in region A and in region B respectively.
- With the discount factor δ , the net present value of 'no reform' payoffs for the entire organization is given by $\frac{R}{1-\delta}$.
- So with a total of i successful reforms in the past in both regions, the net present value of payoffs will be $\frac{(1+i)R}{1-\delta}$.

U-Form and M-Form

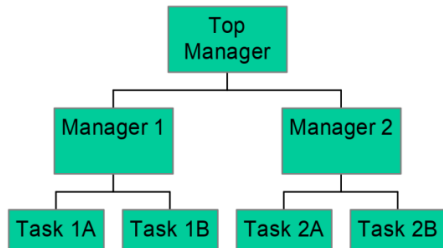


Figure 1: U-Form Organization

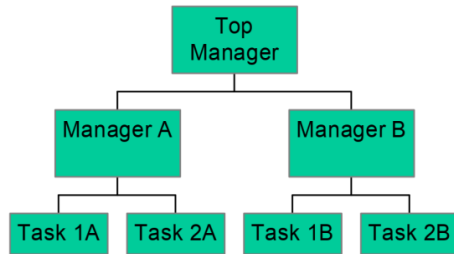


Figure 2: M-Form Organization

Two potential problems

- There is a trade-off between local information and economies of scale, it is related to two potential threats.
- The first problem concerns the quality of its “blueprint.”
- Uncertain outcome: it turns out to be “good” with probability p and “bad” with probability $1-p$.
- A good blueprint for reform raises the payoff from each region permanently by $\frac{R}{2}$ while a bad blueprint always reduces the payoff from each region by $\frac{R}{2}$
- Variable p is stochastically independent
- We assume that task coordination requires a one-time setup cost $\frac{C}{2}$ for each manager.

Two potential problems

- Another problem is about information transmission
- The quality of the coordination depends itself on the quality of communication inside an organization
- Communication problem is endogenous, depending on how tasks are assigned within an organization
- The probability of each message being correct is λ , information transmission is generally imperfect

Two Regions and Two Functions

Full Scale Reform: M-form

- Define stage i as the stage at which a total of i reform programs have been successfully implemented before.
- Let V_i be the net present value of future payoffs at stage i .
- With probability of p , we have $V_{i+1} = \frac{(i+2)R}{1-\delta}$
- With probability of $1 - p$, we have $V_i = \frac{iR}{1-\delta}$
- So $V_i = -C + p[(i+2)R + \delta V_{i+1}] + (1-p)[iR + \delta V_i]$, let $a = \frac{1}{1-(1-p)\delta}$, we have $V_i = -aC + 2paR + aRi + ap\delta V_{i+1}$
- So we can calculate
 $V_0 = -aC \sum_{i=0}^{\infty} (ap\delta)^i + 2paR \sum_{i=0}^{\infty} (ap\delta)^i + aR \sum_{i=0}^{\infty} i(ap\delta)^i$, i.e. under the M-form, the net present value at stage 0 is

$$V_0^{MF} = -\frac{C}{1-\delta} + \frac{pR}{1-\delta} \left(2 + \frac{\delta}{(1-\delta)} \right)$$

Full Scale Reform: U-form

- With probability of $p\lambda^2$, the reform succeeds, we have $V_{i+1} = \frac{(i+2)R}{1-\delta}$
- in other situations, the reform fails, we have $V_i = \frac{iR}{1-\delta}$
- the recursive formula for V_i under the U-form is

$$V_i = -\frac{C}{2} + p\{\lambda^2[(i+2)R + \delta V_{i+1}] + (1-\lambda^2)(iR + \delta V_i)\} + (1-p)[iR + \delta V_i]$$

- Thus under U-form, the net present value at stage 0 is

$$V_0^{UF} = -\frac{C}{2(1-\delta)} + \frac{p\lambda^2 R}{1-\delta} \left(2 + \frac{\delta}{(1-\delta)}\right)$$

- So we get our first conclusion about this comparison: **Proposition 1** *Under full-scale reform, The M-form has a higher net present value than the U-form when the setup cost C is low or the communication quality λ is low, and vice versa.*

Reform with Experiment: M-form

- With probability of p , we have $V_{i+1} = \frac{(i+1)R + (i+2)R}{2(1-\delta)}$
- With probability of $1 - p$, we have $V_i = \frac{iR + (i+1)R}{2(1-\delta)}$, thus we have

$$V_i = -\frac{C}{2} + p\left[\frac{(i+2)R}{2} + \frac{(i+1)R}{2} - \delta\frac{C}{2} + \delta V_{i+1}\right] + (1-p)\left[iR + \frac{(i+1)R}{2} + \delta V_i\right]$$

- Therefore, under M-form with experimentation, the net present value at stage 0 is

$$V_0^{ME} = -\frac{(1+p\delta)C}{2(1-\delta)} + \frac{R}{1-\delta}\left(\frac{1}{2} + \frac{p}{(1-\delta)}\right)$$

- **Proposition 2** *The difference in net present value between the M-form with or without experimentation is*

$$V_0^{ME} - V_0^{MF} = \frac{1}{1-\delta}\left((1-p\delta)\frac{C}{2} - \left(p - \frac{1}{2}\right)R\right)$$

- This is a trade-off between the option value of waiting and the cost of delaying reform in the entire organization.

Difference Between Two Forms

- Under the M-form organization, there are three alternatives: no reform, reform without experimentation, and reform with experimentation
- The overall M-form payoffs are given by

$$V_0^M = \max\left\{\frac{R}{1-\delta}, V_0^{ME}, V_0^{MF}\right\}$$

- Because $\frac{\partial}{\partial p} V_0^{MF} > \frac{\partial}{\partial p} V_0^{ME}$, so let's draw a plot with or without experiment and we get our third conclusions:

Proposition 3 *the M-form with experimentation dominates the M-form without experimentation if and only if $p < p^*$*

Plot of M-form with and without experiment

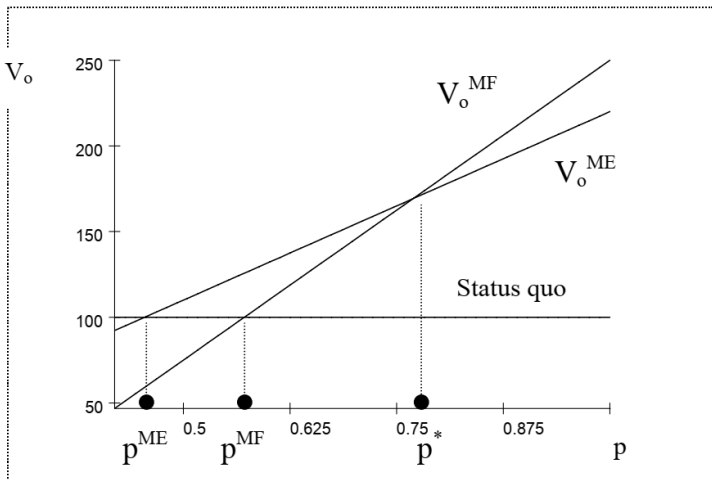


Figure 3: M-form: full scale reform (V_o^{MF}) vs. experimentation (V_o^{ME})

Difference Between Two Forms

- Under a U-form organization, there are only two alternatives: no reform or full-scale reform the overall M-form payoffs are given by

$$V_0^U = \max\left\{\frac{R}{1-\delta}, V_0^{UF}\right\}$$

- Now we get our fourth conclusion:

Proposition 4 *the M-form is better for carrying out reforms when the quality of communication is low, and the M-form with experimentation yields a higher net present value than either the U-form or the M-form without experimentation if in addition the uncertainty of reform blueprint $p < p^*$.*

- We will also draw a plot to show this point.

Plot: M-Form vs U-Form

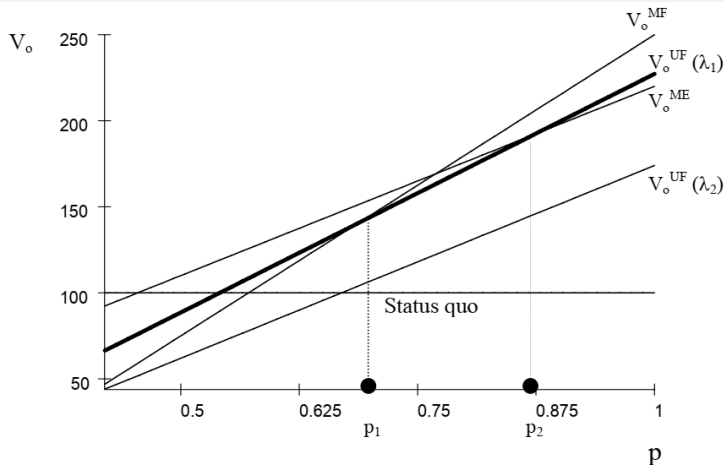


Figure 4: M-form vs. U-form

Generalization

General Situation of M-Form in Experiment

- We now generalize the above model to n regions and m functions.
- We normalize the setup cost of implementing reforms under the M-form to C and the U-form to $\frac{C}{n}$.
- Assume α be the fraction of experimenting regions, so $\alpha \in [\frac{1}{n}, 1]$
- The net present value of payoffs in stage i under the M-form is

$$V_i = -\alpha C + p\{(i+1)R + \alpha R - \delta(1-\alpha)C + \alpha V_{i+1}\} + (1-p)\{(i+1)R - \alpha R + \delta V_i\}$$

- And we get

$$V_0^{M_\alpha} = -\frac{\alpha + (1-\alpha)p\delta}{1-\delta}C + \frac{R}{1-\delta} \left(\alpha(2p-1) + 1 + \frac{p\delta}{1-\alpha} \right)$$

- The next step is to find a suitable α to maximize the result.

Maximize the result

- We let the derivative for α equals to 0, and we can get the result

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$$V_0^{M_\alpha} = \begin{cases} -\frac{C}{1-\delta} + \frac{pR}{1-\delta} \left(2 + \frac{\delta}{1-\delta}\right), & \alpha = 1, p \geq p^* \\ -\frac{C}{1-\delta} \left(\frac{1+(n-1)p\delta}{n}\right) + \frac{R}{1-\delta} \left(\frac{2p+n-1}{n} + \frac{p\delta}{1-\delta}\right) & \alpha = \frac{1}{n}, p < p^* \end{cases}$$

- **Proposition 5**

As long as experimentation is efficient ($p < p^*$), the more regions an economy has the higher the value of experimentation will be.

- Next, we will discuss the general situation with U-Form.

General Situation With U-Form

- The net present value of payoffs in stage i is

$$V_0^U F = -\frac{C}{n} + p\{\lambda^m[(i+2)R + \delta V_i + 1]\} + (1-p)(iR + \delta V_i)$$

- The net present value at stage 0 is

$$V_0^{UF} = -\frac{C}{n(1-\delta)} + \frac{p\lambda^m R}{1-\delta} \left(2 + \frac{\delta}{1-\delta}\right)$$

- An increase in n will increase V_0^{UF} due to the economy of scale advantage while when m increases, the U-form suffers from a disadvantage in coordinating attribute matching.
- Proposition 6** (1) The U-form has a larger positive effect from the increase in the number of regions than the M-form ($\frac{\partial}{\partial n} V_0^{UF} > \frac{\partial}{\partial n} V_0^{M^\alpha} \geq 0$).
(2) The U-form also has a larger negative effect from the increase in the number of functions than the M-form ($\frac{\partial}{\partial m} V_0^{UF} > \frac{\partial}{\partial m} V_0^{M^\alpha} = 0$).

Conditions for the Optimality of M-Form and U-Form

Images of All Types

In theory, many other organizational forms are possible, but we will prove only U and F forms are possible.

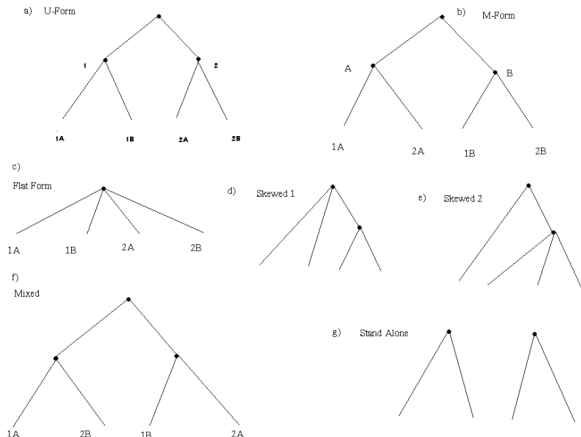


Figure 5: General Organizational Forms.

Violations to assumptions

- Assumption K A manager' s knowledge is limited to one of the following:
 - (K1) functions for any given region;
 - (K2) regions for any given function;
 - (K3) blueprints
- Assume that each manager can acquire only one type of knowledge.
- Assumption K rules out any organizational form employing fewer than 3 managers.
- We also assume costly managers.
- Under these assumptions, only form U and M exist.

Comparisons between Chinese reform and Russia

- Institutional Background of Central Planning in China and the Soviet Union
- In transition economies, there are a greater number of missing markets at the beginning of the reform.
- The Chinese M-form with local government coordinating the various complementary aspects of agricultural reform allowed for flexible and swift coordination (local knowledge and authority) while the Russian government inherited from the Soviet period did not do better in solving coordination problems under central planning.
- Extended Thinking
 - The USA suppress the reform of Russia.
 - The structure of the economy before reform is different, so the motivation is different. Russia needs national force to handle unbalance.
 - Comparative advantage and Winner-Take-All.
 - The lack of investment and talent.

Conclusions

- U-Form: Japan and France, M-Form USA: laboratory of the states
- Compare the performance of organizational forms (M-form and U-form) in implementing changes such as innovation and reform.
- Compared to the U-form, the M-form organization achieves better coordination but suffers from fewer economies of scale.
- The distinctive advantage of the M-form is the flexibility of experimentation, which allows the organization to introduce more innovation and reform.

Thank you!